ML-2 Course Project

Department of Computer Science, Institution Name

Learning Reduced-Round Cipher Behavior using Machine Learning

---

Submitted by

Name	**Roll Number**
Shreyansh Seth	2022484
Prabal Minotra	2022357
Pratham Gautam	2022372
Muthuraj Vairamuth	2022307

March 31, 2026

1 Abstract

This project studies how well common machine learning models can learn reduced-round block and stream ciphers. We implement 16 lightweight ciphers (SIMON, SPECK, PRESENT, PRINCE, TEA, XTEA, RC5, KATAN, RECTANGLE, CHAM, HIGHT, LEA, SIMECK, ChaCha20, Salsa20, and Trivium) and train three models: logistic regression, a multilayer perceptron, and a 1D CNN. For each cipher and round count, we train on randomly sampled plaintexts and evaluate bitwise accuracy and average Hamming distance on a held-out test set. The CNN is the strongest model on average, and meaningful learning typically collapses after a small number of rounds, with an average learnable depth of about 3.67 rounds for the CNN under our thresholds. The results quantify the rapid degradation of learnability as diffusion increases.

2 Introduction

Reduced-round analysis provides a controlled way to probe how quickly a cipher becomes unpredictable. For lightweight primitives with small block sizes and simple round functions, a learning-based distinguisher can sometimes predict ciphertext bits better than chance, but only at shallow round depths. Understanding the point where this advantage disappears is useful for cryptanalysis and for validating diffusion properties in early rounds.

Recent work has shown that neural networks can act as distinguishers for reduced-round block ciphers, notably in the differential cryptanalysis setting [1]. This motivates a broader study that compares multiple ciphers under a shared data generation and training pipeline. Our project focuses on learning the full ciphertext bits directly from plaintext inputs for a fixed secret key and variable round counts.

We implement a suite of 16 ciphers that cover ARX designs, SPN designs, and stream ciphers adapted to the block setting. We train three model families with increasing capacity, compare their accuracy and Hamming distance across rounds, and compute a maximum learnable round $r_{\max}$ using simple accuracy and normalized Hamming thresholds. The key findings are that the CNN generally outperforms the MLP and logistic regression, and that learnability declines quickly with round count for most ciphers.

3 Background

3.1 Block and Stream Ciphers

A reduced-round cipher is modeled as a keyed permutation

$$E_k^r : \{0,1\}^n \to \{0,1\}^n, \tag{1}$$

where n is the block size, k is the secret key, and r is the number of rounds. The block ciphers in this codebase include ARX constructions (SIMON, SPECK, TEA, XTEA, RC5, CHAM, LEA, SIMECK) and S-box based designs (PRESENT, PRINCE, KATAN, RECTANGLE, HIGHT). SIMON and SIMECK use a Simon-like round function with rotations and bitwise AND, while SPECK, TEA, XTEA, RC5, CHAM, and LEA apply combinations of modular addition, rotation, and XOR with round keys. PRESENT and RECTANGLE apply 4-bit S-boxes followed by fixed bit permutations or row shifts; PRINCE uses S-boxes, a linear diffusion layer, and a reflective structure with key whitening; KATAN applies a bit-level nonlinear feedback with fixed round constants; HIGHT uses byte-level F_0/F_1 functions and a simple additive key schedule. Key schedules range from LFSR-like updates (SIMON, SIMECK, KATAN) to round-key rotation and injection (PRESENT, RECTANGLE, PRINCE) and ARX-based updates (LEA, CHAM).

Stream ciphers are adapted as a keystream XOR with plaintext, allowing the model to predict only the message bits. ChaCha20 and Salsa20 use 32-bit add-rotate-xor quarter rounds over a 512-bit state with key and nonce placement, while Trivium uses a 288-bit shift register state with nonlinear feedback and a long initialization phase [14–16]. In the implementation, counter and nonce bits are treated as a header that is copied through, and only the message portion contributes to loss and metrics.

3.2 Machine Learning for Cryptanalysis

Neural distinguishers for reduced-round ciphers have been explored in several works, including Gohr's differential attack on SPECK [1]. These studies show that neural networks can learn statistical biases that disappear as rounds increase. The general approach is to generate labeled plaintext/ciphertext pairs under fixed keys and train a model to predict ciphertext bits or to classify real versus random samples. Our project follows this paradigm but focuses on direct bitwise prediction across multiple ciphers and architectures to quantify the round depth where prediction collapses to chance.

4 Problem Formulation

Given a fixed secret key k and a reduced-round cipher E_k^r, we generate a dataset of random plaintexts $\{x_i\}_{i=1}^N$ and corresponding ciphertexts $y_i = E_k^r(x_i)$. We train a model f_θ to predict the ciphertext bits from plaintext bits:

$$\hat{y}_i = f_\theta(x_i), \quad x_i, y_i \in \{0,1\}^n. \tag{2}$$

We evaluate bitwise accuracy and average Hamming distance over the output bits b (after masking any header bits):

$$\text{Acc} = \frac{1}{Nb} \sum_{i=1}^{N} \sum_{j=1}^{b} \mathbf{1}[\hat{y}_{i,j} = y_{i,j}], \tag{3}$$

$$\text{HD} = \frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{b} \mathbf{1}[\hat{y}_{i,j} \neq y_{i,j}]. \tag{4}$$

Our degradation hypothesis is that, as r increases, learned accuracy approaches 50% and normalized Hamming distance approaches 0.5, indicating indistinguishability from random guessing.

5 Dataset Generation

Plaintexts are sampled uniformly at random from $\{0,1\}^n$, encrypted with a fixed key, and converted into bit vectors. For stream ciphers, counter/nonce or IV bits are treated as header bits and masked out of the loss and metric computations. The dataset is split into 80% train, 10% validation, and 10% test with a fixed random seed. Caches are stored under `data/` using Torch tensors and include metadata for deterministic re-use.

6 Methodology

6.1 Logistic Regression (Baseline)

We use a single linear layer that maps n input bits to n output logits. The model is trained with a masked binary cross-entropy loss to ignore header bits in stream ciphers. This baseline captures linear correlations between plaintext and ciphertext bits.

Algorithm 1 Dataset generation for reduced-round learning

Require: Cipher name C, key k, rounds r, sample count N
Ensure: Train/val/test datasets of bit vectors

```
Initialize cipher instance with key k
Determine block size n and header bits h
for i = 1 to N do
    Sample plaintext x_i ← {0,1}^n
    Compute ciphertext y_i ← E_k^r(x_i)
    Convert x_i, y_i to bit arrays
end for
Build label mask for the n − h output bits
Split into train/val/test with ratio 0.8/0.1/0.1
Optionally cache tensors and metadata to disk
```

6.2 Multi-Layer Perceptron (MLP)

The MLP is a fully connected network with two hidden layers and batch normalization. It captures non-linear bit interactions across the entire block.

6.3 Convolutional Neural Network (CNN)

The CNN uses three parallel 1D convolution branches (kernel sizes 3, 5, and 7) to capture local bit patterns, followed by a fusion layer and two fully connected layers.

6.4 Training Protocol

All models are trained using Adam with learning rate 10^{-3}, batch size 1024, and 15 epochs. The loss is `BCEWithLogitsLoss` with a per-bit mask to exclude header bits. Metrics are computed on the test set using bitwise accuracy and average Hamming distance. The experiment runner applies an early-stop rule: if no model exceeds accuracy 0.55 and normalized Hamming distance 0.45 for two consecutive rounds, it stops increasing r for that cipher.

7 Results & Analysis

Tables 4 and 5 report aggregate results averaged across all ciphers that have metrics for a given round. The number of contributing ciphers decreases at higher rounds because many ciphers stop early; for example, 16 ciphers contribute to $r = 1$ and $r = 2$, 9 to $r = 3$, 5 to $r = 4$, 2 to $r = 5$, and only KATAN contributes for $r \geq 6$.

Table 1: Cipher configurations and maximum tested rounds in the saved results.

Cipher	Type	Block (bits)	Key (bits)	Full rounds	Max tested r
SIMON	Block	32	64	32	4
SPECK	Block	32	64	22	3
PRESENT	Block	64	80	31	4
PRINCE	Block	64	128	12	2
TEA	Block	64	128	32	2
XTEA	Block	64	128	32	2
RC5	Block	64	128	12	2
KATAN	Block	32	80	254	20
RECTANGLE	Block	64	128	25	3
CHAM	Block	128	128	80	5
HIGHT	Block	64	128	32	3
LEA	Block	128	128	24	3
SIMECK	Block	32	64	32	4
ChaCha20	Stream	512	256	20	2
Salsa20	Stream	512	256	20	2
Trivium	Stream	512	80	1152	2

Table 2: MLP architecture.

Layer	Output dim	Activation
Linear	256	ReLU
BatchNorm	256	–
Linear	256	ReLU
BatchNorm	256	–
Linear	n	–

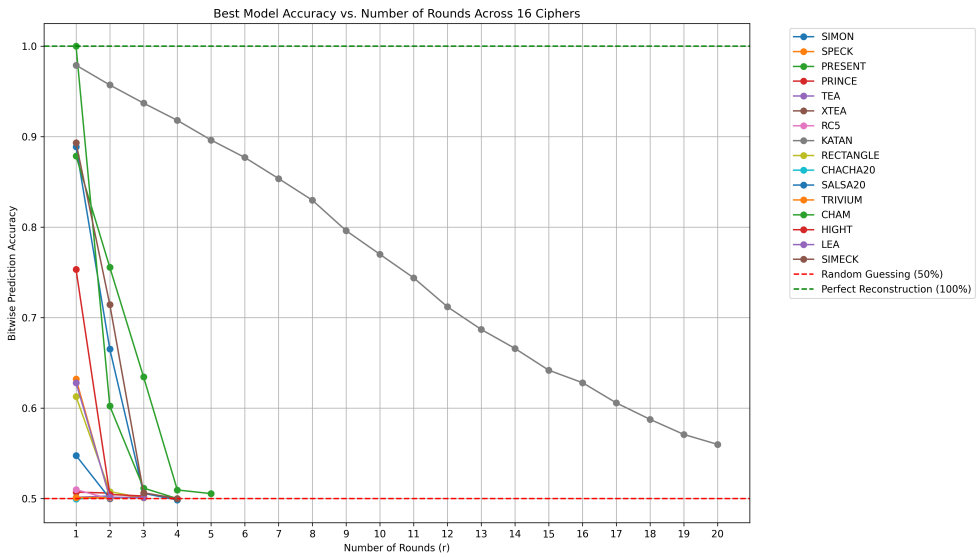

Figure 1: Best-model bitwise accuracy versus round count for each cipher. Each curve selects the highest accuracy among the three models at each round.

Table 3: CNN architecture summary.

Layer	Params
Conv1d branches	$1 \to 32$ channels, kernels 3/5/7, ReLU, BN
Fuse conv	$96 \to 64$ channels, kernel 1, ReLU, BN
FC1	$64n \to 256$, ReLU
FC2	$256 \to n$

Table 4: Average bitwise accuracy (%) by model and round. Values are averaged across available ciphers per round.

Model	r1	r2	r3	r4	r5	r6	r7	r8	r9	r10	r11	r12	r13	r14	r15	r16	r17	r18	r19	r20
Logistic Regression	53.23	51.04	51.58	51.69	54.19	58.07	56.75	56.94	56.00	53.86	53.21	52.82	52.01	53.26	51.53	51.21	50.10	50.41	50.07	49.57
MLP	64.39	55.81	56.15	57.90	68.62	85.19	82.23	80.11	77.12	74.28	70.92	68.20	64.93	63.98	62.19	60.96	59.06	56.98	55.20	54.23
CNN	67.57	57.45	56.53	58.47	70.08	87.69	85.36	82.97	79.60	76.99	74.38	71.18	68.68	66.58	64.18	62.80	60.57	58.74	57.08	55.97

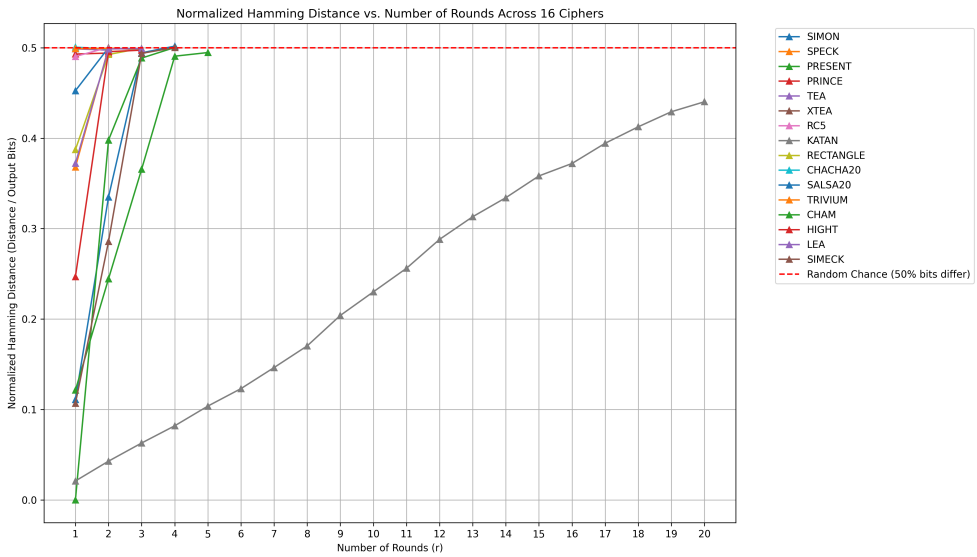

Figure 2: Best-model normalized Hamming distance versus round count for each cipher. Lower values indicate closer reconstruction of ciphertext bits.

Figures 1 and 2 show a consistent degradation trend as rounds increase. The CNN is the strongest model overall (average accuracy 0.634 across all cipher-round pairs), followed by the MLP (0.614) and logistic regression (0.523). Using the Acc ≥ 0.55 and normalized HD ≤ 0.45 criterion, the mean $r_{\max}$ across ciphers with non-null values is 2.50 for logistic regression, 3.44 for the MLP, and 3.67 for the CNN. The plots also reveal that some ciphers (notably KATAN) remain learnable to deeper rounds than others, which is reflected by the higher values in Tables 4 and 5 at large r.

Bonus results. Cross-key generalization results exist for SIMON and KATAN at $r = 5$: the best model is the MLP with accuracy 0.505 for SIMON and 0.834 for KATAN. Partial output learning on SIMON shows strong predictability for the right 16-bit word at $r = 1$ (accuracy 1.0) and $r = 2$ (0.901), while the left word is harder at

Table 5: Average normalized Hamming distance by model and round (HD divided by output bits).

Model	r1	r2	r3	r4	r5	r6	r7	r8	r9	r10	r11	r12	r13	r14	r15	r16	r17	r18	r19	r20
Logistic Regression	0.468	0.490	0.484	0.483	0.458	0.419	0.432	0.431	0.440	0.461	0.468	0.472	0.480	0.467	0.485	0.488	0.499	0.496	0.499	0.504
MLP	0.356	0.442	0.439	0.421	0.314	0.148	0.178	0.199	0.229	0.257	0.291	0.318	0.351	0.360	0.378	0.390	0.409	0.430	0.448	0.458
CNN	0.324	0.425	0.435	0.415	0.299	0.123	0.146	0.170	0.204	0.230	0.256	0.288	0.313	0.334	0.358	0.372	0.394	0.413	0.429	0.440

$r = 2$ (0.507). The distinguisher experiment achieves test accuracy 0.9975 at $r = 1$, dropping to 0.555 at $r = 2$ and 0.487 at $r = 3$.

8 Discussion

Accuracy degrades with increasing rounds because diffusion and avalanche rapidly destroy simple input-output correlations. Early rounds often preserve linear or low-degree relations that the models can exploit, but these correlations vanish as more nonlinear mixing is applied.

Model capacity matters: the CNN can capture local bitwise patterns and combine them through convolutional features, which explains its consistent advantage over the MLP and linear baseline. The MLP still outperforms logistic regression because its nonlinear layers model higher-order interactions across the full block.

The main limitations are the fixed-key setting, the finite sample size (10,000 plaintexts), and early stopping that results in uneven coverage of higher rounds across ciphers. In addition, stream ciphers are treated as blockwise keystream XOR with masked header bits rather than full protocol use. These factors constrain the generality of the conclusions beyond the reduced-round setting.

9 Conclusion

This project evaluates the learnability of reduced-round cipher behavior across 16 lightweight primitives using three standard ML models. The CNN achieves the strongest performance, and the mean learnable depth under our thresholds is below four rounds for most ciphers. These results reinforce the intuition that diffusion quickly eliminates exploitable patterns.

Future work could explore multi-key training, larger datasets, and model architectures specialized for bitwise structure or differential features. Extending the analysis to more rounds with consistent coverage across ciphers would also provide a clearer comparison of long-round behavior.

References

[1] Felix Gohr. Improving Neural Networks for Differential Cryptanalysis. In *CHES*, 2019.

[2] Ray Beaulieu et al. The SIMON and SPECK Families of Lightweight Block Ciphers. *IACR ePrint*, 2015.

[3] Andrey Bogdanov et al. PRESENT: An Ultra-Lightweight Block Cipher. In *CHES*, 2007.

[4] Jens Borghoff et al. PRINCE - A Low-Latency Block Cipher for Pervasive Computing Applications. In *ASIACRYPT*, 2012.

[5] David Wheeler and Roger Needham. TEA, a Tiny Encryption Algorithm. Technical report, 1994.

[6] David Wheeler and Roger Needham. XTEA: eXtended TEA. Technical report, 1997.

[7] Ronald Rivest. The RC5 Encryption Algorithm. In *Fast Software Encryption*, 1994.

[8] Christophe De Canniere, Orr Dunkelman, and Miroslav Knezevic. KATAN and KTANTAN - A Family of Small and Efficient Hardware-Oriented Block Ciphers. In *CHES*, 2009.

[9] Wenling Zhang et al. RECTANGLE: A Bit-slice Lightweight Block Cipher for Low Cost. In *CARDIS*, 2015.

[10] Bonwook Koo et al. CHAM: A Family of Lightweight Block Ciphers for Resource-Constrained Devices. In *ICISC*, 2017.

[11] Deukjo Hong et al. HIGHT: A New Block Cipher Suitable for Low-Resource Device. In *CHES*, 2006.

[12] Deukjo Hong et al. LEA: A 128-bit Block Cipher for Fast Encryption on Common Processors. In *ICISC*, 2013.

[13] Gangqiang Yang et al. The Simeck Family of Lightweight Block Ciphers. In *CHES*, 2015.

[14] Daniel J. Bernstein. ChaCha, a Variant of Salsa20. 2008.

[15] Daniel J. Bernstein. Salsa20 Specification. 2008.

[16] Christophe De Canniere and Bart Preneel. Trivium Specifications. 2006.